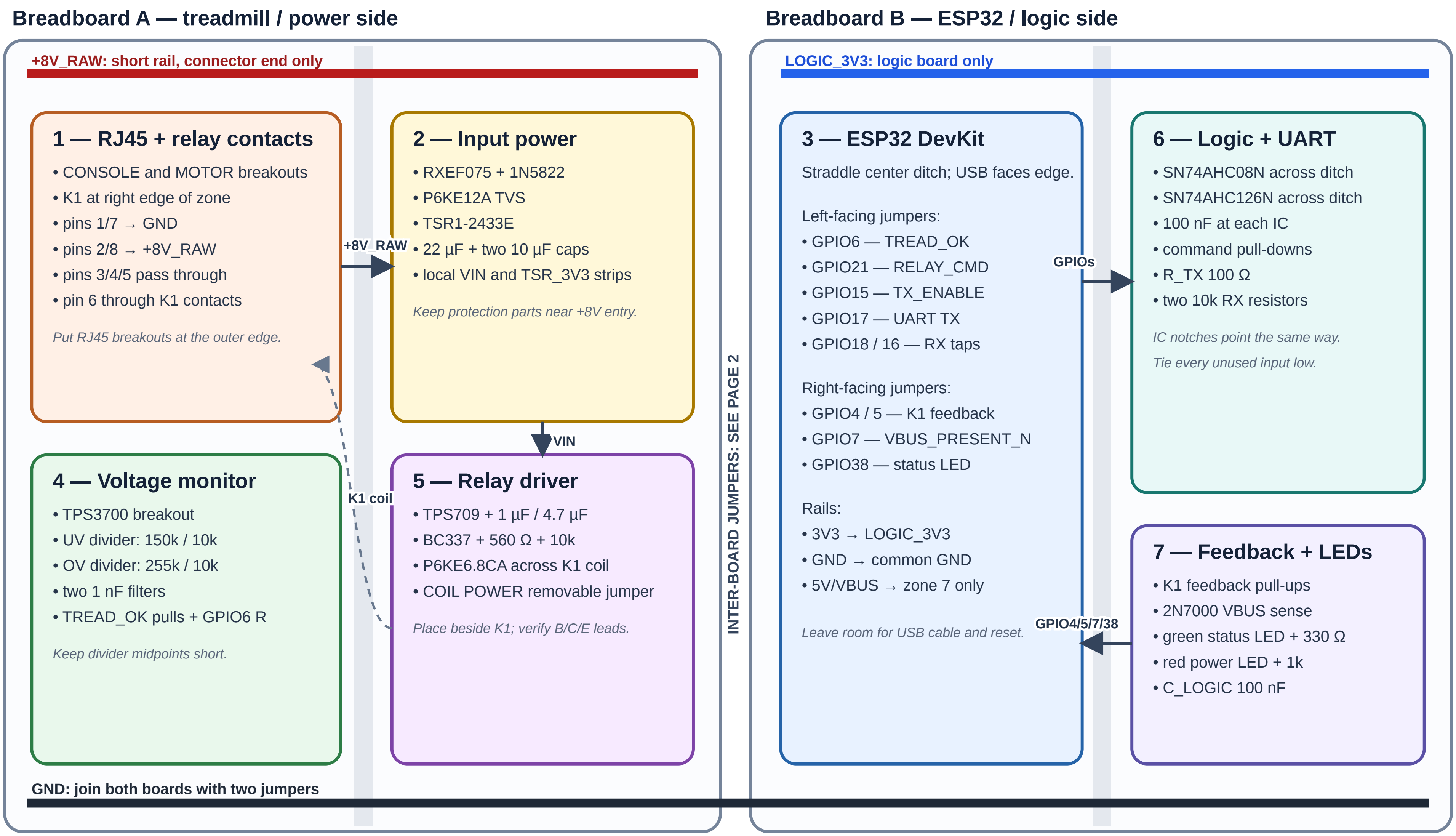
Esp32Tap Breadboard — Logical Cluster Layout

Approximate placement only. Use two full-size breadboards side-by-side; keep each zone compact, then connect zones by the named nets below. Self-contained cluster assembly cards are on pages 3–4.

Start unpowered. Leave STANDALONE POWER and COIL POWER open. Meter-identify relay, transistor, regulator, and breakout pins before inserting them.



Cluster Build Order and Inter-Zone Jumpers

Build and meter-check one zone at a time. Use the point-to-point wiring PDF for every pin-level connection.

Suggested build order

Order	Zone	Stop-and-check point
1	**3 — ESP32**	USB boot works; 3V3 and GND rails are correct.
2	**7 — Feedback/LEDs**	Power LED works; GPIO pins are not shorted.
3	**6 — Logic/UART**	DIP orientation, VCC/GND, unused inputs, and decouplers checked.
4	**1 — RJ45/K1**	Pass-through continuity correct with K1 unpowered.
5	**2 — Input power**	Polarity checked; VIN and GND have no short. Do not join standalone power yet.
6	**4 — Monitor**	Divider values and TREAD_OK pull resistors meter correctly.
7	**5 — Relay driver**	BC337 lead order and coil TVS checked; leave COIL POWER open.

Rail policy

Rail/net	Where	Rule
GND	Both boards	Join boards with two short jumpers; all USB and treadmill grounds are common.
+8V_RAW	Zone 1 → zone 2 only	Do not run along the logic board.
VIN	Zones 2, 4, 5	Keep on treadmill/power board.
LOGIC_3V3	Zones 3, 4, 6, 7	Powered from DevKit USB during initial bring-up.
TSR_3V3	Zone 2	Join to LOGIC_3V3 only through removable STANDALONE POWER jumper.
+5V_RLY	Zone 5	Short local strip only; coil connection stays removable.

Major inter-zone jumpers

Zones	Named nets / purpose
1 → 2	**+8V_RAW, GND** — protected local-power input.
2 → 4	**VIN, GND** — monitor supply and divider source.
2 ⇄ 3	**STANDALONE POWER** — TSR_3V3 to LOGIC_3V3; install only when USB is absent.
3 ⇄ 4	**LOGIC_3V3, TREAD_OK_MCU/GPIO6**.
4 → 6	**TREAD_OK** — hardware permission to both AHC08 gates.
3 → 6	**GPIO21 RELAY_CMD, GPIO15 TX_ENABLE, GPIO17 UART TX**.
6 → 5	**RELAY_GATE** — gated command to TPS709 enable and BC337 base resistor.
5 ⇄ 1	**K1 coil pins 1 and 16** — keep short; COIL POWER remains removable.
6 → 1	**R_TX output to K1 pin 8** — switched ESP transmit.
1 → 3	**CONSOLE pin 6 via 10k → GPIO18; pin-3 net via 10k → GPIO16**.
1 → 7	**K1 pins 9, 11, 13** — NO/COM/NC feedback contacts.
7 ⇄ 3	**GPIO4, GPIO5, GPIO7, GPIO38, 5V/VBUS, LOGIC_3V3**.

Fast physical rules

- Put every DIP IC across a breadboard center ditch, with notches pointing the same direction.
- Put each 100 nF capacitor directly beside its IC supply pins.
- Keep UV_SENSE and OV_SENSE resistor/capacitor nodes inside zone 4.
- Keep BC337, its two resistors, coil TVS, and K1 coil pins together.
- Use colored jumpers consistently: black GND, red raw/VIN, blue 3V3, yellow control, green feedback.
- Label both ends of every jumper that crosses between boards.

This page groups the circuit; it does not replace the 126-row From/To wiring list. Never infer a physical pin from package shape—meter or datasheet-verify it.

Cluster Assembly Cards — Zones 1–4

All order references are from DigiKey invoice **129832848**. Use one card while assembling each cluster. **NOT ORDERED** means absent from that invoice.

1 — RJ45 breakouts + relay contacts

Part	Qty / description	DigiKey part number
K1	1 × G5V-2 DC5 DPDT relay	**39-G5V-2DC5-ND**
RJ45 breakouts	2 × CONSOLE/MOTOR	**NOT ORDERED**

Assemble here: RJ45 pins 1/7 to GND, 2/8 to +8V_RAW, pins 3/4/5 straight through; pin 6 uses K1 pins 4/6/8. Put K1 at the zone 1/5 boundary.

2 — Protected input power

Part	Qty / description	DigiKey part number
F1	1 × RXEF075 resettable fuse	**RXEF075HF-ND**
D1	1 × 1N5822-TP Schottky	**1N5822-TPMSCT-ND**
TVS	1 × P6KE12A-TP	**P6KE12A-TPMSCT-ND**
DC/DC	1 × TSR 1-2433E	**1951-TSR1-2433E-ND**
Bulk cap	1 × 22 µF 25 V aluminum electrolytic, polarized	**732-8821-1-ND**
TSR caps	2 × 10 µF 50 V aluminum electrolytic, polarized	**1189-2322-ND**
Power jumper	1 × removable STANDALONE POWER	**NOT ORDERED**

Assemble here: +8V_RAW → RXEF075 → 1N5822 → VIN. P6KE12A from VIN to GND. Electrolytic body stripe is negative.

3 — ESP32 DevKit

Part	Qty / description	DigiKey part number
DevKit	1 × ESP32-S3-DEVKITC-1-N8R8	**1965-ESP32-S3-DEVKITC-1-N8R8-ND**

Assemble here: straddle the center ditch with USB facing the board edge. DevKit 3V3 feeds LOGIC_3V3 during USB bring-up; DevKit GND joins common GND.

4 — TPS3700 voltage monitor

Part	Qty / description	DigiKey part number
Monitor	1 × TPS3700DDCR	**296-30395-1-ND**
Adapter	1 × LCQT-SOT23-6	**A879AR-ND**
Header	Cut from PRPC040SAAN-RC	**S1011EC-40-ND**
UV top	1 × 150 kΩ 1% axial	**150KXBK-ND**
OV top	1 × 255 kΩ 1% axial	**13-MFR-25FBF52-255K-ND**
10 kΩ	3 × UV/OV bottoms + TREAD_OK pull-up	**NOT ORDERED**
100 kΩ	1 × TREAD_OK pull-down	**NOT ORDERED**
4.7 kΩ	1 × GPIO6 isolation	**NOT ORDERED**
Filter caps	2 × 1 nF 50 V C0G ceramic radial, non-polar	**445-173473-1-ND**
Bypass cap	1 × 100 nF 50 V X7R ceramic radial, non-polar	**BC1084CT-ND**

Assemble here: keep UV_SENSE and OV_SENSE parts close to the adapter; tie TPS3700 outputs together at TREAD_OK.

Cluster Assembly Cards — Zones 5–7

All capacitor construction, polarity, and purchase data is inline. Use equal or higher voltage ratings only.

5 — Relay supply and driver

Part	Qty / description	DigiKey part number
Regulator	1 × TPS70950DBVR	**296-35484-1-ND**
Adapter	1 × LCQT-SOT23-6	**A879AR-ND**
Header	Cut from PRPC040SAAN-RC	**S1011EC-40-ND**
Q relay	1 × BC337-40	**4878-BC337-40CT-ND**
Base resistor	1 × 560 Ω 1% 1/4 W axial	**MFR-25FBF52-560R-ND**
Base pull-down	1 × 10 kΩ	**NOT ORDERED**
Coil TVS	1 × P6KE6.8CA bidirectional	**18-P6KE6.8CACT-ND**
Input cap	1 × 1 µF 25 V X7R ceramic radial, non-polar	**445-173583-1-ND**
Output cap	1 × 4.7 µF 50 V aluminum electrolytic, polarized	**732-8660-1-ND**
Coil jumper	1 × removable COIL POWER	**NOT ORDERED**

Assemble here: verify BC337 B/C/E leads; put TVS directly across K1 pins 1/16. Electrolytic body stripe is negative. Leave COIL POWER open.

6 — Hardware gating and UART

Part	Qty / description	DigiKey part number
AND gates	1 × SN74AHC08N PDIP-14	**296-4524-5-ND**
Buffer	1 × SN74AHC126N PDIP-14	**296-4535-5-ND**
Bypass caps	2 × 100 nF 50 V X7R ceramic radial, non-polar	**BC1084CT-ND**
10 kΩ	4 × command pulls + two RX taps	**NOT ORDERED**
TX resistor	1 × 100 Ω series	**NOT ORDERED**

Assemble here: both ICs straddle the ditch with notches aligned. Put each 100 nF capacitor at pins 14/7. Tie every unused input low.

7 — Relay feedback, VBUS sense, and LEDs

Part	Qty / description	DigiKey part number
VBUS MOSFET	1 × 2N7000	**4878-2N7000CT-ND**
Green LED	1 × LTL-4233	**160-1130-ND**
Green resistor	1 × 330 Ω	**NOT ORDERED**
Red LED	1 × 151051RS11000	**732-5016-ND**
Red resistor	1 × 1 kΩ	**NOT ORDERED**
10 kΩ	4 × feedback/VBUS pulls	**NOT ORDERED**
Logic bypass	1 × 100 nF 50 V X7R ceramic radial, non-polar	**BC1084CT-ND**

Assemble here: meter-identify 2N7000 G/D/S. LED long lead is anode; short/flat-side lead goes toward GND.

Common bench supplies

Item	Quantity / use	DigiKey order status
Full-size breadboards	2, side-by-side	**NOT ORDERED**
Jumper-wire assortment	For 126 From/To connections	**NOT ORDERED**

Color convention: black GND, red raw/VIN, blue 3V3, yellow control, green feedback. Label both ends of every inter-board jumper.

Start unpowered. Leave STANDALONE POWER and COIL POWER removed until their bring-up steps explicitly install them.